Chapter 5: Statistical Inference and Analysis

Master in Computer Science (M1)

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Academic Year 2026–2027

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1 Introduction

Statistical inference is the part of statistics that allows us to draw conclusions about a population using only a sample of observations. In practice, it is often impossible, expensive, or time-consuming to observe all individuals of a population. Instead, we collect data from a smaller group, called a sample, and use this information to estimate unknown quantities or to make decisions.

In data science and machine learning, statistical inference plays a central role. It allows us to quantify uncertainty, evaluate models, compare algorithms, test assumptions, and measure the reliability of predictions. For example, we may want to estimate the average response time of a server, test whether a new algorithm improves prediction accuracy, measure the strength of the relationship between two variables, or evaluate the performance of a regression model.

This chapter begins with the notions of population, sample, statistic, and sampling distribution. These concepts provide the mathematical foundation for estimation and hypothesis testing. We then introduce point estimation and confidence intervals, which are used to estimate unknown parameters such as a mean, a variance, or a proportion.

The second part of the chapter is devoted to hypothesis testing. We explain the logic of statistical tests, the role of the null and alternative hypotheses, and the interpretation of errors and p-values. We distinguish between parametric tests, which rely on distributional assumptions such as normality, and non-parametric tests, which are more flexible when such assumptions are not justified.

Finally, the chapter introduces correlation, covariance, and regression analysis. These tools are fundamental for studying relationships between variables and for building predictive models. We study simple and multiple linear regression, together with model evaluation metrics such as R^2, RMSE, and MAE. Practical sessions in Python will illustrate these ideas using `scipy.stats`, `statsmodels`, and `scikit-learn`.

2 Sampling and Statistical Inference

Statistical inference starts from a simple idea: instead of observing an entire population, we observe only a sample and use it to infer properties of the population. This process requires precise definitions and assumptions, because the quality of the conclusions depends strongly on the way the sample is selected.

2.1 Population and Sample

A **population** is the set of all individuals or objects on which a statistical study is conducted. It is usually denoted by

$$\Omega.$$

If the population is finite, we may write

$$\Omega = \{\omega_1, \omega_2, \ldots, \omega_N\},$$

where N is the size of the population.

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In most real applications, the population is too large to be observed completely. For example, the population may be all users of a platform, all manufactured components produced by a factory, or all patients in a medical database.

A **sample** is a subset of individuals selected from the population. If the sample contains n individuals, we write

$$S = (\omega_1, \omega_2, \ldots, \omega_n).$$

The number n is called the **sample size**.

The objective of statistical inference is to use the information contained in the sample to estimate or test properties of the population.

Example 2.1. *A factory produces metallic rods. The engineer wants to know the average length of all rods produced by the factory.*

Since measuring all rods is impossible, a sample of $n = 50$ *rods is selected. The average length computed from this sample is then used to estimate the true average length of all produced rods.*

2.2 Variable of Interest

In a statistical study, we usually focus on one or several characteristics of the population.

A **variable of interest** is a quantity observed or measured on each individual of the population. It may be quantitative or qualitative.

Examples include:

- the weight of a product;
- the lifetime of a battery;

In statistical inference, the variable of interest is modeled by a random variable X. The population distribution of X is usually unknown, and the goal is to estimate its parameters or test hypotheses about them.

2.3 Random Sample

Let X be a random variable describing the variable of interest in the population. When we observe a sample of size n, we obtain values

$$x_1, x_2, \ldots, x_n.$$

These observed values are viewed as realizations of random variables

$$X_1, X_2, \ldots, X_n.$$

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A **random sample** of size n is a collection of random variables

$$X_1, X_2, \ldots, X_n$$

such that:

- the variables X_i are independent;
- the variables X_i have the same distribution as the population variable X.

We say that $X_1, \ldots, X_n$ are independent and identically distributed, abbreviated as i.i.d.

This assumption is central in classical statistics. It allows us to use probability theory to study the behavior of quantities computed from the sample.

Important Formulas

In classical statistical inference, we usually assume that

$$X_1, X_2, \ldots, X_n$$

are i.i.d. random variables with the same distribution as the population variable X.

2.4 Statistical Inference

Statistical inference is the process of using a sample to draw conclusions about an unknown population.

There are two main objectives:

- **Estimation**: estimating unknown parameters such as a mean, a variance, or a proportion.
- **Hypothesis testing**: deciding whether the observed data are compatible with a proposed assumption about the population.

For example, if μ denotes the unknown mean of a population, we may use the sample mean $\overline{X}$ to estimate μ. If a company claims that the mean lifetime of a product is 1000 hours, we may use a hypothesis test to decide whether the data support or contradict this claim.

3 Statistics and Sampling Distributions

Once a sample is collected, we summarize it using numerical quantities such as the sample mean, sample variance, minimum, maximum, or sample proportion. These quantities are called statistics.

3.1 Statistic

A **statistic** is any function of the sample

$$X_1, X_2, \ldots, X_n.$$

It is usually written as

$$T = f(X_1, X_2, \ldots, X_n).$$

Since a statistic is a function of random variables, it is itself a random variable. Therefore, it has a probability distribution.

The probability distribution of a statistic is called its **sampling distribution**.

The sampling distribution is fundamental because it describes how a statistic varies from one sample to another.

3.2 Sample Mean

The **sample mean** is defined by

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

If

$$E(X_i) = \mu \qquad \text{and} \qquad V(X_i) = \sigma^2,$$

then

$$E(\overline{X}) = \mu \qquad \text{and} \qquad V(\overline{X}) = \frac{\sigma^2}{n}.$$

Important Formulas

The sample mean is centered around the true population mean:

$$E(\overline{X}) = \mu.$$

Its variance decreases when the sample size increases:

$$V(\overline{X}) = \frac{\sigma^2}{n}.$$

Thus, larger samples usually give more stable estimates.

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3.3 Sampling Distribution of the Mean

If the sample size is large, usually

$$n \geq 30,$$

then the Central Limit Theorem gives the approximation

$$\overline{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

Equivalently,

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \approx \mathcal{N}(0,1).$$

Remark. If the population is normally distributed, then the sample mean is normally distributed for every sample size n, not only for large samples.

Example 3.1. *The lifetime of a battery has mean $\mu = 1000$ hours and standard deviation $\sigma = 120$ hours. A sample of $n = 36$ batteries is selected.*

By the Central Limit Theorem,

$$\overline{X} \approx \mathcal{N}\left(1000, \frac{120^2}{36}\right).$$

Thus,

$$\overline{X} \approx \mathcal{N}(1000, 20^2).$$

The probability that the sample mean is greater than 1040 is

$$P(\overline{X} > 1040) = P\left(Z > \frac{1040 - 1000}{20}\right).$$

Hence,

$$P(\overline{X} > 1040) = P(Z > 2) = 1 - \Phi(2).$$

Using the standard normal table,

$$P(\overline{X} > 1040) = 1 - 0.9772 = 0.0228.$$

3.4 Sample Variance

The sample variance measures the dispersion of the observations around the sample mean.

The **empirical variance** is defined by

$$S^2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2.$$

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However, this quantity is biased as an estimator of the population variance σ^2. Therefore, we often use the corrected sample variance.

The **corrected sample variance** is defined by

$$S_c^2 = S^{*2} = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2.$$

The correction $n-1$ instead of n makes S^{*2} an unbiased estimator of σ^2.

Important Formulas

In statistics, when estimating the population variance from a sample, we usually use

$$S^{*2} = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2.$$

3.5 Sampling Distribution of the Variance

If

$$X_1, \ldots, X_n$$

is a random sample from a normal distribution

$$\mathcal{N}(\mu, \sigma^2),$$

then

$$\frac{(n-1)S^{*2}}{\sigma^2} \sim \chi^2(n-1).$$

This result is essential for constructing confidence intervals and hypothesis tests for the variance.

Important Formulas

For a normal population,

$$\frac{(n-1)S^{*2}}{\sigma^2} \sim \chi^2(n-1).$$

3.6 Estimators

Suppose that a random variable X depends on an unknown parameter θ that we wish to estimate.

Examples of parameters frequently encountered in statistics include:

- the population mean μ;

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- the population variance σ^2;
- a proportion p;
- the parameter λ of a Poisson distribution.

Let

$$X_1, X_2, \ldots, X_n$$

be a random sample from the population.

A natural idea is to construct a statistic from the sample and use it to estimate the unknown parameter.

A sequence of random variables

$$\widehat{\theta}_n$$

is called an **estimator** of the parameter θ if

$$\widehat{\theta}_n = f_n(X_1, \ldots, X_n)$$

and

$$\widehat{\theta}_n \xrightarrow{P} \theta \qquad \text{as } n \to \infty.$$

In practice, we often forget that an estimator is formally a sequence of random variables and simply refer to

$$\widehat{\theta}$$

as an estimator of θ.

Example 3.2.
Let

$$X_1, \ldots, X_n$$

be a sample from a population with mean μ. The sample mean

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

is an estimator of μ.

3.7 A Useful Criterion

Directly proving convergence in probability is often difficult.

The following result provides a practical criterion for verifying that a statistic is an estimator.

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Theorem

If

$$E(\widehat{\theta}_n) \longrightarrow \theta$$

and

$$V(\widehat{\theta}_n) \longrightarrow 0$$

when

$$n \to \infty,$$

then

$$\widehat{\theta}_n \xrightarrow{P} \theta.$$

Therefore $\widehat{\theta}_n$ is an estimator of θ.

Example 3.3.

For the sample mean,

$$E(\overline{X}) = \mu$$

and

$$V(\overline{X}) = \frac{\sigma^2}{n} \longrightarrow 0.$$

Therefore,

$$\overline{X} \xrightarrow{P} \mu.$$

Hence $\overline{X}$ is an estimator of the population mean.

4 Properties of Estimators

Several estimators may exist for the same parameter. We therefore need criteria allowing us to compare their quality and select the most appropriate estimator.

4.1 Bias

The estimation error can be written as

$$\widehat{\theta}_n - \theta = \Big(\widehat{\theta}_n - E(\widehat{\theta}_n)\Big) + \Big(E(\widehat{\theta}_n) - \theta\Big).$$

The second term corresponds to a systematic error called the bias.

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The **bias** of an estimator $\widehat{\theta}_n$ is defined by

$$B(\widehat{\theta}_n) = E(\widehat{\theta}_n) - \theta.$$

An estimator is said to be **unbiased** if

$$E(\widehat{\theta}_n) = \theta.$$

Equivalently,

$$B(\widehat{\theta}_n) = 0.$$

An estimator is said to be **asymptotically unbiased** if

$$\lim_{n\to\infty} E(\widehat{\theta}_n) = \theta.$$

Example 4.1.
Since

$$E(\overline{X}) = \mu,$$

the sample mean is an unbiased estimator of the population mean.

Example 4.2.
The empirical variance

$$S^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

satisfies

$$E(S^2) = \frac{n-1}{n}\sigma^2.$$

Therefore,

$$B(S^2) = -\frac{\sigma^2}{n}.$$

Hence S^2 is a biased estimator of σ^2.

4.2 Corrected Sample Variance

The empirical variance

$$S^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

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is a biased estimator of the population variance σ^2, since

$$E(S^2) = \frac{n-1}{n}\sigma^2.$$

To correct this bias, we define the corrected sample variance.

The **corrected sample variance** is defined by

$$S_c^2 = \frac{n}{n-1}S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2.$$

Important Formulas

The corrected sample variance satisfies

$$E(S_c^2) = \sigma^2.$$

Therefore, S_c^2 is an unbiased estimator of the population variance.

4.3 Mean Squared Error

The bias measures the systematic error of an estimator, while the variance measures its fluctuations from one sample to another. In practice, both quantities must be considered together.

The **Mean Squared Error** of an estimator $\widehat{\theta}_n$ of a parameter θ is defined by

$$MSE(\widehat{\theta}_n) = E\left[(\widehat{\theta}_n - \theta)^2\right].$$

The Mean Squared Error can be decomposed as

$$MSE(\widehat{\theta}_n) = V(\widehat{\theta}_n) + \left[E(\widehat{\theta}_n) - \theta\right]^2.$$

Important Formulas

$$MSE(\widehat{\theta}_n) = V(\widehat{\theta}_n) + B(\widehat{\theta}_n)^2.$$

A good estimator should have both a small variance and a small bias.

Among unbiased estimators, the most precise estimator is generally the one with the smallest variance.

5 Maximum Likelihood Estimation

Maximum likelihood estimation is one of the most important methods for constructing estimators. It is widely used in statistics, data science, and machine learning.

The main idea is simple: we choose the value of the unknown parameter that makes the observed sample the most likely.

Let

$$x_1, x_2, \ldots, x_n$$

be observed values from a random sample

$$X_1, X_2, \ldots, X_n.$$

Assume that the distribution of X depends on an unknown parameter θ.

The **likelihood function** is the function of θ defined from the observed sample. If X is discrete, then

$$L(\theta) = \prod_{i=1}^{n} P(X_i = x_i).$$

If X is continuous with density $f(x;\theta)$, then

$$L(\theta) = \prod_{i=1}^{n} f(x_i;\theta).$$

The maximum likelihood estimator is obtained by maximizing the likelihood function.

The **maximum likelihood estimator** of θ is the value

$$\widehat{\theta}$$

that maximizes

$$L(\theta).$$

Equivalently, since the logarithm is an increasing function, we often maximize the log-likelihood

$$\ell(\theta) = \ln L(\theta).$$

Important Formulas

In practice, maximum likelihood estimation is usually done by solving

$$\frac{d}{d\theta}\ell(\theta) = 0.$$

Example 5.1. *Let*

$$X_1, \ldots, X_n$$

be a random sample from a Poisson distribution with parameter λ:

$$X_i \sim \mathcal{P}(\lambda).$$

The probability mass function is

$$P(X_i = x_i) = e^{-\lambda}\frac{\lambda^{x_i}}{x_i!}.$$

The likelihood function is

$$L(\lambda) = \prod_{i=1}^{n} e^{-\lambda}\frac{\lambda^{x_i}}{x_i!}.$$

Thus,

$$L(\lambda) = e^{-n\lambda}\prod_{i=1}^{n}\frac{\lambda^{x_i}}{x_i!}.$$

The log-likelihood is

$$\ell(\lambda) = -n\lambda + \left(\sum_{i=1}^{n} x_i\right)\ln(\lambda) - \sum_{i=1}^{n}\ln(x_i!).$$

We differentiate with respect to λ:

$$\frac{d}{d\lambda}\ell(\lambda) = -n + \frac{1}{\lambda}\sum_{i=1}^{n} x_i.$$

Solving

$$\frac{d}{d\lambda}\ell(\lambda) = 0$$

gives

$$-n + \frac{1}{\lambda}\sum_{i=1}^{n} x_i = 0.$$

Therefore,

$$\widehat{\lambda} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

Hence, the maximum likelihood estimator of λ is

$$\widehat{\lambda} = \overline{X}.$$

Example 5.2. *Let*

$$X_1, \ldots, X_n$$

be a random sample from a normal distribution

$$\mathcal{N}(\mu, \sigma^2),$$

where μ is known and σ is unknown.
The density is

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$$f(x_i;\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right).$$

The likelihood function is

$$L(\sigma) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right).$$

Therefore,

$$L(\sigma) = (2\pi)^{-n/2}\sigma^{-n} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right).$$

The log-likelihood is

$$\ell(\sigma) = -\frac{n}{2}\ln(2\pi) - n\ln(\sigma) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2.$$

By maximizing this function, we obtain

$$\widehat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i-\mu)^2}.$$

Thus, the maximum likelihood estimator of σ is

$$\widehat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i-\mu)^2}.$$

6 Confidence Intervals

A point estimator provides a single numerical value for an unknown parameter. However, because the estimator is computed from a sample, it is subject to random fluctuations.

For this reason, it is often preferable to provide an interval that is likely to contain the true value of the parameter rather than a single number.

Such intervals are called confidence intervals.

6.1 General Principle

Let θ be an unknown parameter.

Suppose that two statistics

$$L(X_1,\ldots,X_n)$$

and

$$U(X_1,\ldots,X_n)$$

can be constructed such that

$$P\left(L \leq \theta \leq U\right) = 1-\alpha.$$

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Then

$$[L, U]$$

is called a confidence interval for θ with confidence level

$$1 - \alpha.$$

A confidence interval of level

$$1 - \alpha$$

for a parameter θ is a random interval

$$[L, U]$$

satisfying

$$P(L \leq \theta \leq U) = 1 - \alpha.$$

The quantity

$$1 - \alpha$$

is called the confidence level.
Common values are

$$90\%, \quad 95\%, \quad 99\%.$$

Important Formulas

For a confidence level of 95%,

$$\alpha = 0.05.$$

For a confidence level of 99%,

$$\alpha = 0.01.$$

Remark.

A confidence level of 95% does not mean that the probability that the parameter belongs to the computed interval is 95%.

It means that if we repeated the sampling procedure many times, about 95% of the constructed intervals would contain the true parameter.

6.2 Confidence Interval for a Mean with Known Variance

Suppose that

$$X_1, X_2, \ldots, X_n$$

is a random sample from a population with mean μ and variance σ^2, where σ is known.

According to the Central Limit Theorem, for large samples (or for any sample size if the population is normal),

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1).$$

Let $z_{\alpha/2}$ denote the quantile of order

$$1 - \frac{\alpha}{2}$$

of the standard normal distribution.

Then

$$P\left(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}\right) = 1 - \alpha.$$

Replacing Z by its expression gives

$$P\left(-z_{\alpha/2} \leq \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha.$$

Multiplying by

$$\frac{\sigma}{\sqrt{n}}$$

yields

$$P\left(-z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \overline{X} - \mu \leq z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

Finally,

$$P\left(\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

Important Formulas

The confidence interval of level $1 - \alpha$ for the mean μ, when the variance is known, is

$$IC_{1-\alpha}(\mu) = \left[\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right].$$

Common values of $z_{\alpha/2}$ are:

$$z_{0.05} = 1.645, \qquad z_{0.025} = 1.96, \qquad z_{0.005} = 2.576.$$

These values correspond respectively to confidence levels

$$90\%, \qquad 95\%, \qquad 99\%.$$

Example 6.1.

A machine produces metallic rods.

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The length of the rods has known standard deviation

$$\sigma = 4 \ cm.$$

A sample of

$$n = 64$$

rods is selected and the sample mean is

$$\overline{x} = 102 \ cm.$$

Construct a 95% *confidence interval for the true mean length.*
For a confidence level of 95%,

$$z_{0.025} = 1.96.$$

The margin of error is

$$1.96 \times \frac{4}{\sqrt{64}} = 1.96 \times 0.5 = 0.98.$$

Therefore,

$$IC_{95\%} = [102 - 0.98, \ 102 + 0.98].$$

Hence,

$$IC_{95\%} = [101.02, \ 102.98].$$

We are therefore 95% *confident that the true mean length belongs to this interval.*

6.3 Confidence Interval for a Mean with Unknown Variance

In practice, the population variance σ^2 is rarely known.

When σ is unknown, it is replaced by the sample standard deviation

$$S_c = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2}.$$

If the population follows a normal distribution, then the statistic

$$T = \frac{\overline{X} - \mu}{S_c/\sqrt{n}}$$

follows a Student distribution with

$$n - 1$$

degrees of freedom.
We write

$$T \sim t(n-1).$$

Let

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$$t_{\alpha/2,n-1}$$

denote the quantile of order

$$1-\frac{\alpha}{2}$$

of the Student distribution with $n-1$ degrees of freedom.
Then

$$P\left(-t_{\alpha/2,n-1} \leq T \leq t_{\alpha/2,n-1}\right) = 1-\alpha.$$

Replacing T by its expression yields

$$P\left(-t_{\alpha/2,n-1} \leq \frac{\overline{X}-\mu}{S_c/\sqrt{n}} \leq t_{\alpha/2,n-1}\right) = 1-\alpha.$$

After rearranging the inequality, we obtain

$$P\left(\overline{X} - t_{\alpha/2,n-1}\frac{S_c}{\sqrt{n}} \leq \mu \leq \overline{X} + t_{\alpha/2,n-1}\frac{S_c}{\sqrt{n}}\right) = 1-\alpha.$$

Important Formulas

When the variance is unknown, a confidence interval of level $1-\alpha$ for the population mean is

$$IC_{1-\alpha}(\mu) = \left[\overline{X} - t_{\alpha/2,n-1}\frac{S_c}{\sqrt{n}},\ \overline{X} + t_{\alpha/2,n-1}\frac{S_c}{\sqrt{n}}\right].$$

Remark.
For large sample sizes,

$$t_{\alpha/2,n-1} \approx z_{\alpha/2}.$$

Therefore, the Student distribution approaches the standard normal distribution when n becomes large.

Example 6.2.
A sample of

$$n = 16$$

students is selected.
The observed average score is

$$\overline{x} = 72.$$

The corrected sample standard deviation is

$$S_c = 8.$$

Construct a 95% confidence interval for the population mean.
Since

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$$n = 16,$$

the number of degrees of freedom is

$$15.$$

From the Student table,

$$t_{0.025,15} = 2.131.$$

The margin of error is

$$2.131 \times \frac{8}{\sqrt{16}} = 2.131 \times 2 = 4.262.$$

Therefore,

$$IC_{95\%} = [72 - 4.262,\ 72 + 4.262].$$

Hence,

$$IC_{95\%} = [67.74,\ 76.26].$$

We conclude that the true mean score is estimated to lie between 67.74 *and* 76.26 *with a confidence level of* 95%*.*

6.4 Confidence Interval for a Proportion

Many practical problems involve estimating a proportion.

Examples include:

- the proportion of defective products;
- the proportion of customers satisfied with a service;
- the proportion of patients responding positively to a treatment.

Let

$$p$$

denote the unknown population proportion. Let

$$X$$

be the number of individuals possessing the characteristic of interest in a sample of size n. Then

$$X \sim \mathcal{B}(n, p).$$

The sample proportion is

$$\widehat{p} = \frac{X}{n}.$$

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For sufficiently large samples,

$$\widehat{p} \approx \mathcal{N}\left(p, \frac{p(1-p)}{n}\right).$$

Consequently,

$$\frac{\widehat{p}-p}{\sqrt{\frac{p(1-p)}{n}}} \approx \mathcal{N}(0,1).$$

Replacing the unknown quantity p by $\widehat{p}$, we obtain the following confidence interval.

Important Formulas

A confidence interval of level $1-\alpha$ for a proportion p is

$$IC_{1-\alpha}(p) = \left[\widehat{p} - z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}},\ \widehat{p} + z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}\right].$$

Example 6.3.

Among

$$200$$

customers surveyed, 120 *declared that they were satisfied with a service. The estimated proportion is*

$$\widehat{p} = \frac{120}{200} = 0.60.$$

For a confidence level of 95%,

$$z_{0.025} = 1.96.$$

The margin of error is

$$1.96\sqrt{\frac{0.6(1-0.6)}{200}}.$$

Hence,

$$1.96\sqrt{\frac{0.24}{200}} = 0.0679.$$

Therefore,

$$IC_{95\%} = [0.60 - 0.0679,\ 0.60 + 0.0679].$$

Thus,

$$IC_{95\%} = [0.532,\ 0.668].$$

We estimate that the true proportion of satisfied customers lies between 53.2% *and* 66.8%.

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6.5 Confidence Interval for a Variance

In some applications, the variability of a population is as important as its mean.

For example, in industrial quality control, a large variance may indicate an unstable production process even if the mean remains unchanged.

Suppose that

$$X_1, X_2, \ldots, X_n$$

is a random sample from a normal distribution

$$\mathcal{N}(\mu, \sigma^2).$$

We have seen that

$$\frac{(n-1)S_c^2}{\sigma^2} \sim \chi^2(n-1),$$

where S_c^2 denotes the corrected sample variance.

Let

$$\chi^2_{\alpha/2,n-1}$$

and

$$\chi^2_{1-\alpha/2,n-1}$$

be the corresponding quantiles of the Chi-square distribution.

Then

$$P\left(\chi^2_{\alpha/2,n-1} \leq \frac{(n-1)S_c^2}{\sigma^2} \leq \chi^2_{1-\alpha/2,n-1}\right) = 1-\alpha.$$

After rearranging the inequality, we obtain

$$P\left(\frac{(n-1)S_c^2}{\chi^2_{1-\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)S_c^2}{\chi^2_{\alpha/2,n-1}}\right) = 1-\alpha.$$

Important Formulas

A confidence interval of level $1-\alpha$ for the population variance σ^2 is

$$IC_{1-\alpha}(\sigma^2) = \left[\frac{(n-1)S_c^2}{\chi^2_{1-\alpha/2,n-1}}, \frac{(n-1)S_c^2}{\chi^2_{\alpha/2,n-1}}\right].$$

If a confidence interval for the standard deviation is required, we simply take the square root of both bounds.

$$IC_{1-\alpha}(\sigma) = \left[\sqrt{\frac{(n-1)S_c^2}{\chi^2_{1-\alpha/2,n-1}}}, \sqrt{\frac{(n-1)S_c^2}{\chi^2_{\alpha/2,n-1}}}\right].$$

Example 6.4.

A sample of

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$$n = 20$$

observations is collected from a normal population.
The corrected sample variance is

$$S_c^2 = 25.$$

Construct a 95% *confidence interval for the population variance.*
Since

$$n - 1 = 19,$$

the Chi-square table gives approximately

$$\chi^2_{0.975,19} = 32.852, \qquad \chi^2_{0.025,19} = 8.907.$$

Therefore,

$$IC_{95\%}(\sigma^2) = \left[\frac{19 \times 25}{32.852},\ \frac{19 \times 25}{8.907}\right].$$

Hence,

$$IC_{95\%}(\sigma^2) = [14.46,\ 53.33].$$

We conclude that the population variance is estimated to lie between

$$14.46$$

and

$$53.33$$

with a confidence level of 95%.

7 Hypothesis Testing

Confidence intervals provide a range of plausible values for an unknown parameter. Another common objective of statistical inference is to decide whether a claim about a population is supported by the observed data.

This decision-making framework is called hypothesis testing.

Hypothesis testing is widely used in science, engineering, medicine, economics, and machine learning. For example, we may wish to determine:

- whether a new algorithm performs better than an existing one;
- whether the mean lifetime of a product exceeds a required value;
- whether two populations have the same mean;
- whether a treatment has a significant effect.

The general principle is to compare the observed data with a reference hypothesis and determine whether the discrepancy is large enough to reject that hypothesis.

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7.1 Statistical Hypotheses

A **statistical hypothesis** is a statement concerning one or more parameters of a population.

Two hypotheses are considered:

- the **null hypothesis**, denoted H_0;
- the **alternative hypothesis**, denoted H_1.

The null hypothesis generally represents the current belief or the reference situation, while the alternative hypothesis represents the claim that we seek to establish.

Example 7.1.
Suppose that a manufacturer claims that the average lifetime of a battery is

$$1000$$

hours.
We may test

$$H_0 : \mu = 1000$$

against

$$H_1 : \mu \neq 1000.$$

7.2 Type I Error and Type II Error

When performing a statistical test, the decision is based on a sample rather than on the entire population. Consequently, there is always a risk of making an incorrect decision.

Two types of errors may occur.

A **Type I Error** occurs when the null hypothesis H_0 is true but is rejected.
The probability of committing a Type I Error is denoted by

$$\alpha.$$

A **Type II Error** occurs when the null hypothesis H_0 is false but is not rejected.
The probability of committing a Type II Error is denoted by

$$\beta.$$

The following table summarizes the possible situations.

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	Accept H_0	Reject H_0
H_0 true	Correct decision	Type I Error
H_0 false	Type II Error	Correct decision

Important Formulas

The probability of a Type I Error is

$$\alpha.$$

The probability of a Type II Error is

$$\beta.$$

7.3 Power of a Statistical Test

A good statistical test should be able to detect false null hypotheses.

The **power** of a statistical test is defined by

$$1 - \beta.$$

The power represents the probability of correctly rejecting a false null hypothesis. A powerful test has a large value of

$$1 - \beta.$$

and therefore a small probability of Type II Error.

Important Formulas

$$\text{Power} = 1 - \beta.$$

A large power indicates a high probability of detecting a real effect.

7.4 General Procedure for Hypothesis Testing

Most statistical tests follow the same sequence of steps.

1. State the null hypothesis H_0.
2. State the alternative hypothesis H_1.
3. Choose the significance level α.
4. Compute an appropriate test statistic.
5. Determine the rejection region or compute the p-value.
6. Draw a conclusion.

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Example 7.2.
Suppose that a company claims that the average lifetime of a battery is

$$1000$$

hours.
A researcher suspects that the true average lifetime is smaller.
The hypotheses are

$$H_0 : \mu = 1000$$

and

$$H_1 : \mu < 1000.$$

After collecting a sample, a suitable test statistic is computed and compared with a critical value obtained from the corresponding probability distribution.

7.5 p-Value

Modern statistical software generally reports a quantity called the p-value.

The p-**value** is the probability, assuming that H_0 is true, of obtaining a result at least as extreme as the one observed.

The smaller the p-value, the stronger the evidence against the null hypothesis.
The decision rule is simple.

Important Formulas

If

$$p\text{-value} < \alpha,$$

we reject H_0.
If

$$p\text{-value} \geq \alpha,$$

we do not reject H_0.

Remark.
Failing to reject H_0 does not prove that H_0 is true.
It simply means that the available data do not provide sufficient evidence against H_0.

7.6 One-Sided and Two-Sided Tests

The choice of the alternative hypothesis determines the type of test.

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- Left-tailed test:

$$H_0 : \mu = \mu_0, \qquad H_1 : \mu < \mu_0.$$

- Right-tailed test:

$$H_0 : \mu = \mu_0, \qquad H_1 : \mu > \mu_0.$$

- Two-sided test:

$$H_0 : \mu = \mu_0, \qquad H_1 : \mu \neq \mu_0.$$

The rejection region depends on the chosen alternative hypothesis.

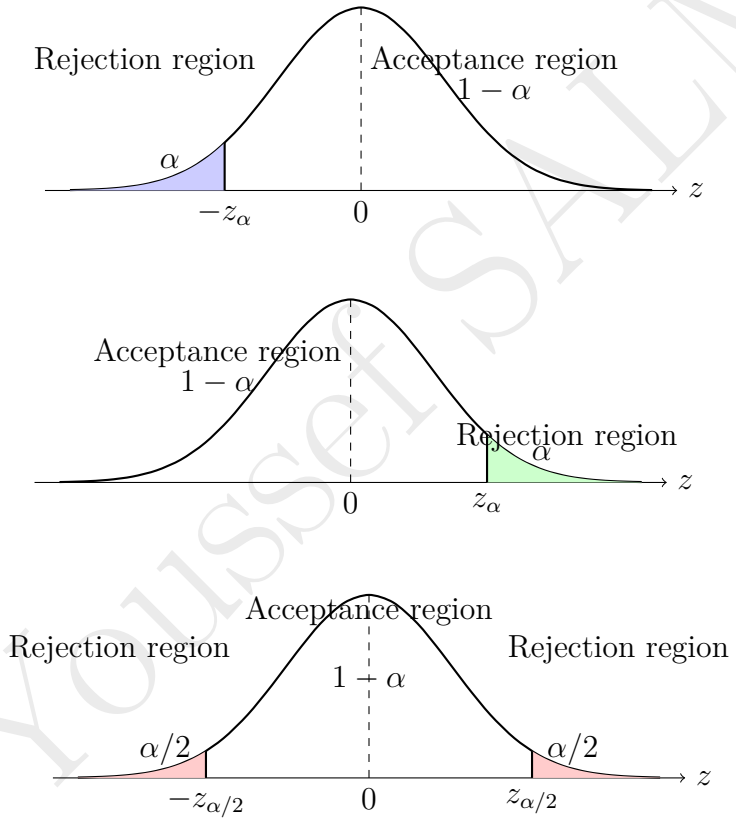


Figure 1: Rejection regions for left-tailed, right-tailed, and two-tailed hypothesis tests.

8 Parametric Tests

Parametric tests are statistical tests that assume that the population follows a specific probability distribution, most often the normal distribution.

These tests are generally more powerful than non-parametric tests when their assumptions are satisfied.

The most commonly used parametric tests are:

- the Z-test;

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- the Student t-test;
- the Chi-square test.

8.1 Z-Test for a Mean

The Z-test is used to test a hypothesis concerning a population mean when the population variance is known.

Suppose that

$$X_1, \ldots, X_n$$

is a random sample from a population with mean μ and known variance σ^2.
We wish to test

$$H_0 : \mu = \mu_0$$

against one of the alternatives

$$H_1 : \mu < \mu_0, \qquad H_1 : \mu > \mu_0, \qquad H_1 : \mu \neq \mu_0.$$

Under the null hypothesis,

$$Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1).$$

Important Formulas

The Z-test statistic is

$$Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}.$$

Under H_0,

$$Z \sim \mathcal{N}(0,1).$$

Example 8.1.

A machine produces screws whose length is known to have standard deviation

$$\sigma = 2 \ mm.$$

The manufacturer claims that the average length is

$$10 \ mm.$$

A sample of

$$n = 25$$

screws is selected and gives

$$\overline{x} = 9.2.$$

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Test

$$H_0 : \mu = 10$$

against

$$H_1 : \mu < 10$$

at the significance level

$$\alpha = 5\%.$$

The test statistic is

$$Z = \frac{9.2 - 10}{2/\sqrt{25}} = \frac{-0.8}{0.4} = -2.$$

The critical value is

$$-z_{0.05} = -1.645.$$

Since

$$-2 < -1.645,$$

the statistic belongs to the rejection region.
Therefore we reject H_0.
There is sufficient evidence to conclude that the mean length is smaller than 10 *mm.*

8.2 Student t-Test for a Mean

In practice, the population variance σ^2 is rarely known.

When the variance is unknown, it is replaced by the corrected sample variance

$$S_c^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2.$$

If the population is normally distributed, then under the null hypothesis

$$H_0 : \mu = \mu_0,$$

the statistic

$$T = \frac{\overline{X} - \mu_0}{S_c/\sqrt{n}}$$

follows a Student distribution with

$$n - 1$$

degrees of freedom.
We write

$$T \sim t(n-1).$$

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Important Formulas

The Student test statistic is

$$T = \frac{\overline{X} - \mu_0}{S_c/\sqrt{n}}.$$

Under H_0,

$$T \sim t(n-1).$$

The decision rule is similar to that of the Z-test, except that the critical values are obtained from the Student distribution table.

Example 8.2.

A university claims that students spend on average

$$20$$

hours per week studying.
A sample of

$$n = 16$$

students gives

$$\overline{x} = 23$$

hours and

$$S_c = 4.$$

We wish to test

$$H_0 : \mu = 20$$

against

$$H_1 : \mu > 20$$

at the significance level

$$\alpha = 5\%.$$

The test statistic is

$$T = \frac{23 - 20}{4/\sqrt{16}} = \frac{3}{1} = 3.$$

Since

$$n - 1 = 15,$$

the critical value is

$$t_{0.05,15} = 1.753.$$

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Because

$$3 > 1.753,$$

the statistic belongs to the rejection region.

Therefore, we reject H_0.

There is sufficient evidence to conclude that students spend more than 20 hours per week studying.

8.3 Comparison Between the Z-Test and the Student Test

The Z-test and Student test are very similar. The main difference lies in the knowledge of the population variance.

Test	Variance	Distribution
Z-test	Known	$\mathcal{N}(0,1)$
Student test	Unknown	$t(n-1)$

Important Formulas

Use the Z-test when the population variance σ^2 is known.
Use the Student test when the population variance is unknown.
In practice, the Student test is used much more frequently.

8.4 Chi-Square Test for a Variance

Sometimes the objective is not to test a mean but rather the variability of a population.

Suppose that

$$X_1, \ldots, X_n$$

is a random sample from a normal population.

We wish to test

$$H_0 : \sigma^2 = \sigma_0^2.$$

Under H_0,

$$\chi^2 = \frac{(n-1)S_c^2}{\sigma_0^2}$$

follows a Chi-square distribution with

$$n - 1$$

degrees of freedom.

$$\chi^2 \sim \chi^2(n-1).$$

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Important Formulas

The Chi-square test statistic is

$$\chi^2 = \frac{(n-1)S_c^2}{\sigma_0^2}.$$

Under H_0,

$$\chi^2 \sim \chi^2(n-1).$$

Example 8.3.

A manufacturer claims that the variance of the diameter of produced components is

$$\sigma^2 = 4.$$

A sample of

$$n = 20$$

components gives

$$S_c^2 = 6.$$

The test statistic is

$$\chi^2 = \frac{19 \times 6}{4} = 28.5.$$

This value is then compared with the appropriate Chi-square critical values obtained from the table.

9 Non-Parametric Tests

The parametric tests presented previously rely on assumptions about the population distribution, most often normality.

In some situations, these assumptions are not satisfied or cannot be verified. In such cases, non-parametric tests provide an alternative approach.

Non-parametric tests generally require fewer assumptions and can be used with ordinal data, small samples, or non-normal distributions.

Important Formulas

Parametric tests assume a specific probability distribution.
Non-parametric tests require fewer assumptions and are therefore more robust when the distribution is unknown.

9.1 Sign Test

The sign test is one of the simplest non-parametric tests.

It is used to test hypotheses concerning a population median.

Suppose that we wish to test

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$$H_0 : m = m_0,$$

where m denotes the population median.

For each observation x_i, we compare x_i with m_0.

$$\begin{cases} + & \text{if } x_i > m_0, \\ - & \text{if } x_i < m_0. \end{cases}$$

Under the null hypothesis, positive and negative signs should occur with the same probability.

Therefore, the number of positive signs follows approximately a binomial distribution

$$B(n, 0.5).$$

Example 9.1.

A company claims that the median delivery time is

3

days.

A sample of customers is observed and each delivery time is compared with 3 *days.*

If a significantly larger number of observations exceeds 3 *days, the claim may be rejected.*

9.2 Mann–Whitney Test

The Mann–Whitney test is a non-parametric alternative to the two-sample Student test.

It is used to compare two independent populations without assuming normality.

The observations from both samples are ranked together from the smallest to the largest value.

The test statistic is then computed from the ranks rather than from the raw observations.

Example 9.2.

Suppose that two machine learning algorithms are evaluated on several datasets.

If the performance scores are not normally distributed, the Mann–Whitney test can be used to determine whether one algorithm tends to produce better results than the other.

9.3 Comparison Between Parametric and Non-Parametric Tests

Property	Parametric	Non-Parametric
Distribution assumption	Required	Not required
Statistical power	Higher	Lower
Robustness	Lower	Higher
Typical examples	Z-test, t-test	Sign test, Mann–Whitney

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Important Formulas

When the assumptions of parametric tests are satisfied, parametric methods are generally preferred because they provide greater statistical power.
When the assumptions are violated, non-parametric methods often provide more reliable conclusions.

10 Simple Linear Regression

10.1 Introduction

In many applications, we seek to understand how one variable influences another.
Examples include:

- predicting house prices from surface area;
- predicting energy consumption from temperature;
- predicting exam scores from study time;
- predicting sales from advertising expenditure.

Regression analysis provides a mathematical framework for modeling such relationships.

The simplest regression model involves one explanatory variable and one response variable.

10.2 The Simple Linear Regression Model

Let

$$Y$$

denote the response (dependent) variable and

$$X$$

the explanatory (independent) variable.
The simple linear regression model is

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \qquad i = 1, \ldots, n.$$

where

- β_0 is the intercept;
- β_1 is the slope;
- ε_i is the random error term.

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A simple linear regression model is a statistical model describing a linear relationship between a response variable Y and a single predictor variable X.

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i.$$

The parameter β_1 measures the expected change in Y when X increases by one unit. The parameter β_0 represents the expected value of Y when

$$X = 0.$$

Example 10.1. *Suppose that*

$$Y = \textit{Exam Score}$$

and

$$X = \textit{Hours of Study.}$$

A fitted model

$$\widehat{Y} = 40 + 5X$$

means that:

- *a student studying* 0 *hours is expected to obtain a score of* 40*;*
- *each additional hour of study increases the expected score by* 5 *points.*

10.3 Ordinary Least Squares

In practice, the parameters

$$\beta_0 \qquad \text{and} \qquad \beta_1$$

are unknown. They must be estimated from the observed data

$$(x_1, y_1), \ldots, (x_n, y_n).$$

The fitted regression line is written

$$\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i.$$

The difference between the observed value y_i and the fitted value $\widehat{y}_i$ is called the residual.

The **residual** associated with observation i is

$$\widehat{\varepsilon}_i = y_i - \widehat{y}_i.$$

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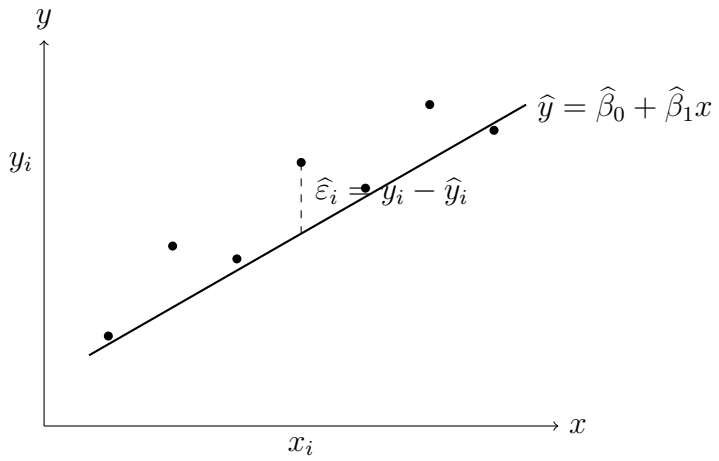


Figure 2: Observed value, fitted value, and residual in simple linear regression.

The Ordinary Least Squares method consists of choosing $\widehat{\beta}_0$ and $\widehat{\beta}_1$ in order to minimize the sum of squared residuals:

$$\sum_{i=1}^{n} \widehat{\varepsilon}_i^2 = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2 .$$

Since

$$\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i,$$

we minimize

$$\sum_{i=1}^{n} \left(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i\right)^2 .$$

Important Formulas

The Ordinary Least Squares method minimizes

$$\phi(\widehat{\beta}_0, \widehat{\beta}_1) = \sum_{i=1}^{n} \left(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i\right)^2 .$$

10.3.1 Derivation of the OLS Estimators

We want to minimize

$$\phi(\widehat{\beta}_0, \widehat{\beta}_1) = \sum_{i=1}^{n} \left(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i\right)^2 .$$

The minimum is obtained by solving the system

$$\frac{\partial \phi}{\partial \widehat{\beta}_0} = 0 \qquad \text{and} \qquad \frac{\partial \phi}{\partial \widehat{\beta}_1} = 0.$$

First,

$$\frac{\partial \phi}{\partial \widehat{\beta}_0} = -2 \sum_{i=1}^{n} \left(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i\right).$$

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Thus,

$$\frac{\partial \phi}{\partial \widehat{\beta}_0} = 0$$

gives

$$\sum_{i=1}^{n} \left(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i \right) = 0.$$

Hence,

$$\sum_{i=1}^{n} y_i - n\widehat{\beta}_0 - \widehat{\beta}_1 \sum_{i=1}^{n} x_i = 0.$$

Dividing by n, we obtain

$$\overline{y} - \widehat{\beta}_0 - \widehat{\beta}_1 \overline{x} = 0.$$

Therefore,

$$\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}.$$

This shows that the regression line passes through the point

$$(\overline{x}, \overline{y}).$$

Now,

$$\frac{\partial \phi}{\partial \widehat{\beta}_1} = -2 \sum_{i=1}^{n} x_i \left(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i \right).$$

Thus,

$$\frac{\partial \phi}{\partial \widehat{\beta}_1} = 0$$

gives

$$\sum_{i=1}^{n} x_i \left(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i \right) = 0.$$

Replacing

$$\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}$$

we get

$$\sum_{i=1}^{n} x_i \left(y_i - \overline{y} + \widehat{\beta}_1 \overline{x} - \widehat{\beta}_1 x_i \right) = 0.$$

Therefore,

$$\sum_{i=1}^{n} x_i (y_i - \overline{y}) - \widehat{\beta}_1 \sum_{i=1}^{n} x_i (x_i - \overline{x}) = 0.$$

Hence,

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$$\widehat{\beta}_1 = \frac{\sum_{i=1}^n x_i(y_i - \overline{y})}{\sum_{i=1}^n x_i(x_i - \overline{x})}.$$

Now, since

$$\sum_{i=1}^n (y_i - \overline{y}) = 0,$$

we have

$$\sum_{i=1}^n x_i(y_i - \overline{y}) = \sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y}).$$

Also,

$$\sum_{i=1}^n x_i(x_i - \overline{x}) = \sum_{i=1}^n (x_i - \overline{x})^2.$$

Thus,

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^n (x_i - \overline{x})^2}.$$

Finally,

$$\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}.$$

Important Formulas

For simple linear regression,

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^n (x_i - \overline{x})^2}$$

and

$$\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}.$$

The fitted regression model is therefore

$$\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x.$$

Example 10.2. *Consider the following data:*

x	1	2	3	4	5
y	2	3	5	4	6

We have

$$\overline{x} = 3$$

and

$$\overline{y} = 4.$$

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Moreover,

$$\sum_{i=1}^{5}(x_i - \overline{x})(y_i - \overline{y}) = 9$$

and

$$\sum_{i=1}^{5}(x_i - \overline{x})^2 = 10.$$

Thus,

$$\widehat{\beta}_1 = \frac{9}{10} = 0.9.$$

Then

$$\widehat{\beta}_0 = 4 - 0.9 \times 3 = 1.3.$$

The fitted regression line is

$$\widehat{y} = 1.3 + 0.9x.$$

10.4 Coefficient of Determination (R^2)

After fitting a regression model, an important question arises:

How well does the regression line explain the observed data?

The coefficient of determination, denoted by

$$R^2,$$

measures the proportion of variability in the response variable that is explained by the regression model.

To define R^2, we introduce three quantities.

10.4.1 Total Sum of Squares

The total variability of the response variable is measured by

$$SST = \sum_{i=1}^{n}(y_i - \overline{y})^2.$$

10.4.2 Regression Sum of Squares

The variability explained by the model is

$$SSR = \sum_{i=1}^{n}(\widehat{y}_i - \overline{y})^2.$$

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10.4.3 Residual Sum of Squares

The unexplained variability is

$$SSE = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2.$$

These quantities satisfy

$$SST = SSR + SSE.$$

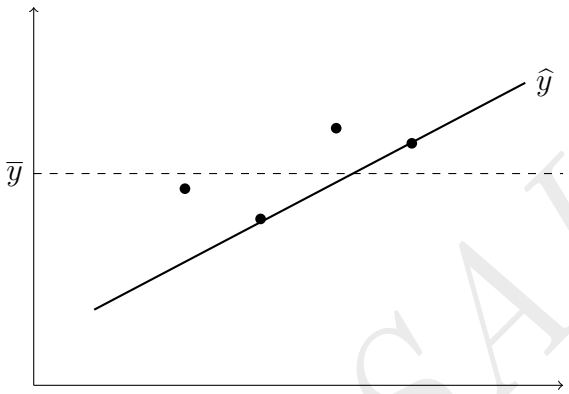


Figure 3: Observed values, mean response, and fitted regression line.

The coefficient of determination is

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}.$$

Important Formulas

$$R^2 = 1 - \frac{SSE}{SST}.$$

$$0 \leq R^2 \leq 1.$$

Interpretation:

- $R^2 = 0$: the model explains none of the variability.
- $R^2 = 1$: the model perfectly explains the data.
- Larger values indicate a better fit.

Example 10.3.
Suppose that

$$SST = 120$$

and

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$$SSE = 30.$$

Then

$$R^2 = 1 - \frac{30}{120} = 0.75.$$

The model explains

$$75\%$$

of the variability of the response variable.

11 Model Evaluation Metrics

Once a regression model has been fitted, it is important to evaluate its prediction accuracy.

Several performance metrics are commonly used in statistics, data science, and machine learning.

The most frequently used metrics are:

- Mean Squared Error (MSE);
- Root Mean Squared Error (RMSE);
- Mean Absolute Error (MAE);
- Coefficient of Determination (R^2).

Suppose that

$$y_1, \ldots, y_n$$

are the observed values and

$$\widehat{y}_1, \ldots, \widehat{y}_n$$

are the predicted values.

The prediction error for observation i is

$$e_i = y_i - \widehat{y}_i.$$

11.1 Mean Squared Error (MSE)

The Mean Squared Error is defined by

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2.$$

Important Formulas

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2.$$

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The MSE penalizes large errors more strongly because the errors are squared.
A smaller MSE indicates a better model.

Example 11.1.
Suppose that the observed values are

$$(10, 12, 15)$$

and the predicted values are

$$(11, 11, 14).$$

Then

$$MSE = \frac{(10-11)^2 + (12-11)^2 + (15-14)^2}{3} = \frac{3}{3} = 1.$$

11.2 Root Mean Squared Error (RMSE)

The RMSE is simply the square root of the MSE.

$$RMSE = \sqrt{MSE}.$$

Important Formulas

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}.$$

Unlike the MSE, the RMSE is expressed in the same unit as the response variable.
For this reason, RMSE is often easier to interpret.

Example 11.2.
If

$$MSE = 9,$$

then

$$RMSE = \sqrt{9} = 3.$$

The average prediction error is therefore approximately three units.

11.3 Mean Absolute Error (MAE)

Another commonly used metric is the Mean Absolute Error.
Instead of squaring the errors, we take their absolute values.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y}_i|.$$

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Important Formulas

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y}_i|.$$

The MAE is less sensitive to extreme observations than the RMSE.

Example 11.3.

Using the previous observations,

$$MAE = \frac{|10-11| + |12-11| + |15-14|}{3} = \frac{3}{3} = 1.$$

11.4 Comparison Between MSE, RMSE and MAE

Metric	Formula	Interpretation
MSE	$\frac{1}{n}\sum e_i^2$	Penalizes large errors
RMSE	$\sqrt{\frac{1}{n}\sum e_i^2}$	Same unit as the response
MAE	$\frac{1}{n}\sum \lvert e_i \rvert$	Average absolute error

Important Formulas

A good predictive model should have

$$R^2$$

close to 1,
and

$$MSE, \quad RMSE, \quad MAE$$

as small as possible.

12 Multiple Linear Regression

Simple linear regression studies the relationship between a response variable and a single explanatory variable.

However, in many real-world applications, the response depends on several explanatory variables simultaneously.

For example:

- House prices may depend on surface area, number of rooms, age of the building, and location.
- Electricity consumption may depend on temperature, humidity, and the number of occupants.
- Student performance may depend on study time, attendance, and previous academic results.

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- The accuracy of a machine learning model may depend on several hyperparameters.

To model such situations, we use multiple linear regression.

12.1 Multiple Linear Regression Model

Let

$$Y$$

be the response variable and

$$X_1, X_2, \ldots, X_p$$

be p explanatory variables.
The multiple linear regression model is

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip} + \varepsilon_i, \qquad i = 1, \ldots, n.$$

where

- β_0 is the intercept;
- β_j measures the effect of variable X_j;
- ε_i is the random error term.

A **multiple linear regression model** relates a response variable to several explanatory variables through a linear equation

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon.$$

12.2 Interpretation of the Coefficients

Each coefficient β_j represents the expected change in the response variable when X_j increases by one unit while all other variables remain fixed.

Example 12.1.
Consider the model

$$\widehat{Y} = 20 + 3X_1 + 5X_2,$$

where

- X_1 = *number of study hours;*
- X_2 = *number of completed assignments.*

The coefficient 3 *indicates that an additional hour of study increases the expected score by approximately three points, assuming the number of assignments remains unchanged.*

The coefficient 5 *indicates that one additional completed assignment increases the expected score by approximately five points, assuming the study time remains unchanged.*

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12.3 Matrix Formulation

Multiple linear regression is often written in matrix form.

Let

$$Y = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \qquad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}.$$

Define the design matrix

$$X = \begin{pmatrix} 1 & X_{11} & X_{12} & \cdots & X_{1p} \\ 1 & X_{21} & X_{22} & \cdots & X_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \cdots & X_{np} \end{pmatrix}.$$

The model becomes

$$Y = X\beta + \varepsilon.$$

This compact notation is widely used in machine learning and numerical computing.

12.4 Ordinary Least Squares Estimation

As in simple linear regression, the parameters are estimated by minimizing the sum of squared residuals.

The least-squares estimator is

$$\widehat{\beta} = (X^T X)^{-1} X^T Y.$$

Important Formulas

For multiple linear regression,

$$\widehat{\beta} = (X^T X)^{-1} X^T Y.$$

This formula provides the estimates of all regression coefficients simultaneously.

12.5 Geometrical Interpretation

The fitted model attempts to find the hyperplane that best approximates the observed data.

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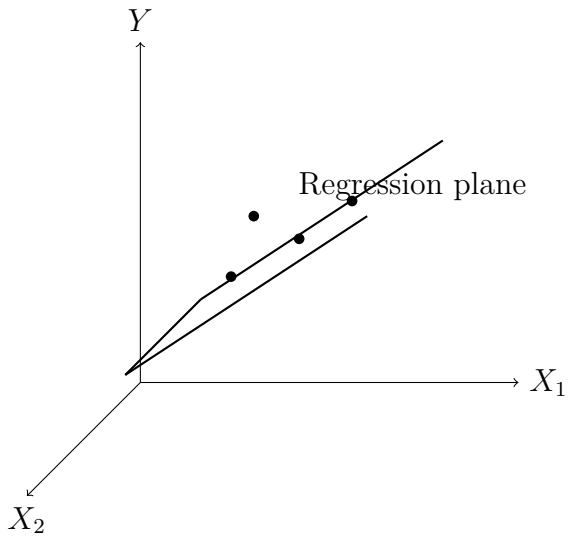


Figure 4: Conceptual representation of a regression plane in multiple linear regression.

12.6 Advantages of Multiple Regression

Multiple regression offers several advantages:

- It can model more realistic situations involving several variables.
- It generally provides better predictions than simple regression.
- It allows the effect of each variable to be studied while keeping the others fixed.
- It is the foundation of many machine learning algorithms.

12.7 From Statistics to Machine Learning

Many supervised machine learning algorithms can be viewed as extensions of multiple linear regression.

Examples include:

- Ridge Regression;
- Lasso Regression;
- Logistic Regression;
- Neural Networks.

For this reason, multiple linear regression occupies a central position between classical statistics and modern machine learning.

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Important Formulas

Simple Linear Regression:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Multiple Linear Regression:

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon.$$

The multiple model generalizes the simple model by incorporating several predictors simultaneously.

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